Class 20: The Registrar's Report

Survivorship bias, from Wald's bombers to your diploma

Team Members: ______________________________ Date: ______________

PART 1: BEFORE WE LOOK AT ANYTHING ELSE

Every year, a college can report a reassuring number: the average time it takes a student to earn their degree. Here is that number for one entering cohort, straight from the registrar's records – 63 students who have graduated.

> **Q1.1. Before you run anything: does a number like this – computed only from students who already graduated – describe the typical experience of an entering student? Is it likely to be optimistic, pessimistic, or accurate? And roughly what fraction of the entering cohort do you think this statistic is even built from?**

Caution: Write an answer above before you run the next cell. The rest of today only works if you commit first.

PART 2: WHAT THE REPORT LEFT OUT

The registrar's report only ever tracks students who finished. Here is the **entire** entering cohort those 63 graduates came from – everyone who started, including the ones who dropped out and the ones still working on it.

> **Q2.1. Run the cell that computes the honest completion rate. Record it here, and compare it to your guess in Q1.1.**

Honest completion rate: ____________ My Q1.1 guess was: ____________

> **Q2.2 Discussion. The naive graduates-only average (4.31 years) and this honest completion rate are two completely different kinds of claim. What question does the average answer, and what question does it fail to answer at all?**

PART 3: THE PROPER WAY TO COUNT

Averaging only the finishers has a second, sneakier problem: even the students who are **on track** to graduate eventually don't count yet if we ask today. Statisticians call this **censoring** – we know how long we've tracked a student, but not whether or when they'll graduate. There's a standard tool built for exactly this situation – a **Kaplan-Meier curve**, marked with a tick wherever a censored student leaves the count – and your instructor will walk through one built from scratch, using only the Table methods and numpy you already know.

> **Q3.1. At year 6 – the edge of our tracking window – what percentage of the original entering cohort still has not graduated? Is this number knowable from the registrar's graduates-only report?**

> **Q3.2. The curve puts the point where 50% of the entering cohort has graduated at 4.5 years – later than the naive graduates-only average of 4.31 years. Which direction did the naive number get it wrong, and why does that direction make sense once you know what it was leaving out?**

PART 4: BACK TO THE BOMBERS

In lecture: engineers wanted to armor the planes where returning bombers showed the most damage. Wald's insight was that the returning planes were exactly the wrong place to look.

Q4.1. Fill in the right-hand column, matching each row to today's activity.

	Returning Bombers	**The Registrar's Report**
What we can see	Damage on planes that made it back	______________________ ______________________
What's missing from the data	Planes that never came back	______________________ ______________________
The naive conclusion	Armor where the damage is	______________________ ______________________
Why it's wrong	Damage there was survivable – that's **why** those planes made it back	______________________ ______________________

Q4.2. Name one more place – outside planes and college – where only looking at the "survivors" would give you a misleading picture.